Sophia Tsang

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Education

Cornell University, College of Engineering **Expected May 2027**
Bachelor of Science in Computer Science, Minor in Robotics **GPA: 3.923**
Relevant Courses: Frontiers of Computer Vision, Intro to ML, Foundations of Robotics, Robot Learning, Robot Perception, Intro to Game Architecture, Analysis of Algorithms, Formal Verification, Differential Equations, Discrete Mathematics

Experience

MIT Lincoln Laboratory — Lexington, MA

Software Engineering and Robotics Intern | Python, PyTorch, TensorFlow, Docker, ROS2, VIO **May 2026 – Aug 2026**

- Built an end-to-end 3D terrain reconstruction pipeline for UAV imagery, recovering 3D surface geometry from *limited-perspective aerial views in real time* and achieving **92.0% geometric accuracy** against LiDAR ground-truth terrain
- Adapted *EOGS satellite photogrammetry* approach to UAV imagery, combining RPC camera modeling with VIO-based sun estimation to incorporate *physically accurate shadow geometry* as an additional geometric cue for Gaussian splatting
- Optimized the real-time processing pipeline by leveraging multiprocessing to decouple ROS2 camera stream dataset updates from background depth estimation, enabling *non-blocking data ingestion and continuous, live 3D map generation*

Cornell Electric Vehicles — Ithaca, NY

Perception Lead, Autonomy Engineer | Rust, Python, C++, PyTorch, ROS2, Docker **Oct 2024 – Present**

- Built a *Rust LiDAR driver* with a calibrated sensor measurement model, decoding raw UDP packets and applying range, angular, timing, and sensor-offset corrections to generate 3D point clouds for *live obstacle detection*
- Developed C++ and Rust implementations of *real-time two-layer-graph clustering* on **80k+ points**, reducing LiDAR scan latency **10**× to **20–30 ms** for dynamic obstacle region generation and detection, tracking, and longer-horizon planning
- Implemented a *real-time* point cloud to costmap algorithm that constructs accurate representations of free-space and obstacle regions around the vehicle, *enabling accurate and continuous spatial mapping* for path planning

Noah Snavely Research Group — Roosevelt Island, NY

Research Intern | PyTorch, COLMAP, CUDA, Gemini, GCP, Flask, Render, React, Three.js, WebGL **June 2025 – Jan 2026**

- Co-authored ECCV 2026 paper, "Honey, I Shrunk the Arc de Triomphe", on *metric-scale monocular geometry estimation*
- Developed the *geo-referencing pipeline* that scales SfM reconstructions to absolute metric scale by leveraging geo-tagged imagery, optimizing viewpoint selection to improve large-scale 3D scene reconstruction
- Expanded Doppelgangers++ into a vision-language network that performs *end-to-end 3D reconstruction*, designed to visually disambiguate between distinct, yet look-alike surfaces and serve as a *more robust alternative to classical SfM*
- Built a web viewer that runs on React, WebGL, and Flask to render **4k+ doppelganger-pruned 3D reconstructions**

Projects

Re:Color | Java, LibGDX, Agile Project Management **Jan 2026 – May 2026**

- Designed a color-based gameplay system with 7 other developers, enabling players to recolor the world by painting and absorbing color from the 2D platformer environment, and shipped the game *live* at GDIAC Showcase to **400+ attendees**
- Developed core player systems by implementing a behavior tree for *player state management*, synchronizing animations and refining movement physics, and building a *custom paint splatter rendering engine*

3D Desktop Simulation | Python, Flask, C#, Unity, React, Three.js, WebGL, REST, WebSocket, GCP **July 2025 – Present**

- Built an interactive 3D desktop environment using Three.js for *real-time 3D rendering*, simulating a computer interface with custom GLSL fragment shaders and draggable project apps displayed in popup browser windows

Skills

Languages: Python, C/C++, C#, Java, Rocq, OCaml, JavaScript/TypeScript, SQL, HTML, CSS, Verilog, Rust
Frameworks/Libraries: PyTorch, NumPy, Pandas, React.js, Three.js, Flask, OpenCV, sqlite3, WebGL, Matplotlib
Tools: Git, CI/CD, ROS2, Linux, Docker, VSCode, LibGDX, Unity, Render, GCP, Figma, Fusion 360, Slurm